IDEATION PHASE

Define the Problem Statements

Date	13 February 2026
Team ID	
Project Name	Malaria Detection using Deep Learning
Maximum marks	2

Problem Statements

Laboratory technicians and healthcare professionals involved in malaria diagnosis face significant challenges due to the time-consuming and manual nature of microscopic blood smear analysis. The increasing patient load, especially during malaria outbreaks, combined with limited laboratory resources and a shortage of skilled personnel, often results in diagnostic delays and increased risk of human error. These challenges create pressure to maintain high accuracy while delivering timely results, directly impacting patient safety and treatment decisions. Therefore, there is a critical need for a reliable, efficient, and user-friendly automated diagnostic support system that enhances accuracy, reduces workload, and assists healthcare professionals in making faster and more consistent decisions.

Element	**PS-1**	**PS-2**
I am (Customer)	Laboratory technician in diagnostic laboratory	Healthcare professional handling malaria cases
I'm trying to	Provide fast and accurate diagnosis	Ensure timely and reliable treatment
But	Manual microscopy is time-consuming	Heavy workload increases errors
Because	Diagnosis depends on human observation	Limited access to automated tools
Which makes me feel	Stressed under tight deadlines	Anxious about patient safety

Table 1 (a) Customer Problem Statement